

Research and Documentation

Accessing Software from Outside the Local Network using Python Environment and Open WebUI with ngrok\

1. Objective

The objective of this report is to study how locally hosted software can be accessed from outside the local network. In this setup, a Python environment is used to run Open WebUI, and ngrok is used to expose the local server to the internet securely.

2. Introduction

When software runs on a local machine (localhost), it can only be accessed within the same system. To access it externally, the local server must be exposed using tunneling tools like ngrok or through network configurations such as port forwarding.

In this project, Open WebUI is hosted on a Python environment and made accessible outside the local network using ngrok.

3. Tools and Technologies Used

Python 3.10+
Virtual Environment (venv)
Open WebUI
OpenRouter API
ngrok
Windows Operating System

4. System Setup (Python Environment)

Step 1: Create Virtual Environment

```
python -m venv openwebui-env
```

Step 2: Activate Virtual Environment

Windows:

```
openwebui-env\Scripts\activate
```

Step 3: Install Open WebUI

```
pip install open-webui
```

5. Running Open WebUI Locally

Start the Server

```
open-webui serve
```

Access in Browser

<http://localhost:8080>

At this stage, the application is only accessible locally.

```
C:\Windows\System32\cmd.exe
Microsoft Windows [Version 10.0.26200.8246]
(c) Microsoft Corporation. All rights reserved.

C:\Users\91831\OneDrive\Documents\WINOpen>.\.venv\Scripts\Activate

(.venv) C:\Users\91831\OneDrive\Documents\WINOpen>where python
C:\Users\91831\OneDrive\Documents\WINOpen\venv\Scripts\python.exe
C:\Users\91831\pyenv\pyenv-win\shims\python
C:\Users\91831\pyenv\pyenv-win\shims\python.bat
C:\Users\91831\AppData\Local\Microsoft\WindowsApps\python.exe

(.venv) C:\Users\91831\OneDrive\Documents\WINOpen>python -c "import sys; print(sys.executable)"
C:\Users\91831\OneDrive\Documents\WINOpen\venv\Scripts\python.exe

(.venv) C:\Users\91831\OneDrive\Documents\WINOpen>ngrok version
ngrok version 3.39.1-msix-stable

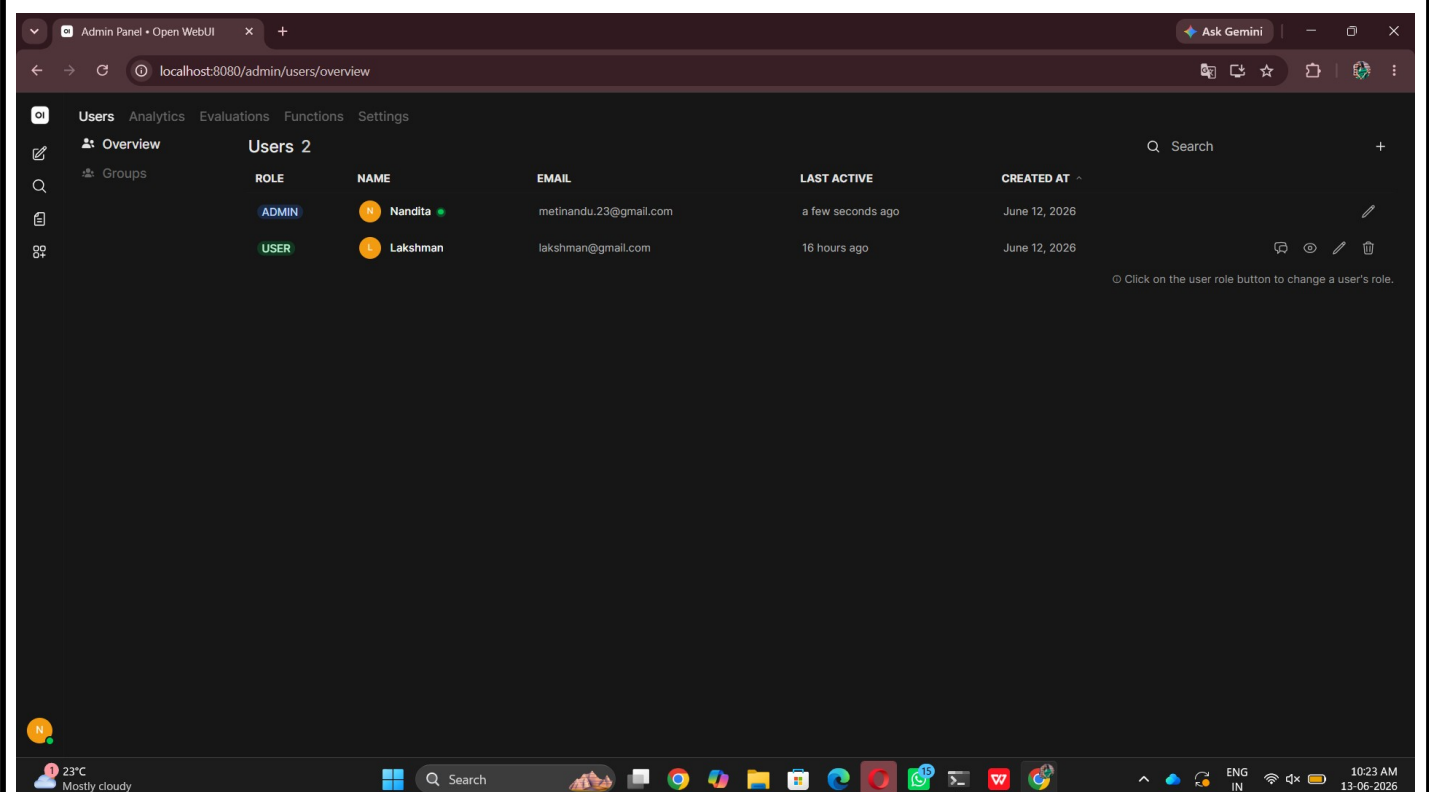
(.venv) C:\Users\91831\OneDrive\Documents\WINOpen>ngrok config add-authtoken 3F7IF3exs7MD9ajbZH1SaAXUVCO_7RT4ebodPADqfoHrzyDnJ
Authtoken saved to configuration file: C:\Users\91831\AppData\Local\ngrok\ngrok.yml

(.venv) C:\Users\91831\OneDrive\Documents\WINOpen>open-webui serve
Loading WEBUI_SECRET_KEY from file, not provided as an environment variable.
Loading WEBUI_SECRET_KEY from C:\Users\91831\OneDrive\Documents\WINOpen\.webui_secret_key
INFO [alembic.runtime.migration] Context impl SQLiteImpl.
INFO [alembic.runtime.migration] Will assume non-transactional DDL.
WARNI [open_webui.env]

WARNING: CORS_ALLOW_ORIGIN IS SET TO '*' - NOT RECOMMENDED FOR PRODUCTION DEPLOYMENTS.

C:\Users\91831\OneDrive\Documents\WINOpen\venv\Lib\site-packages\pydub\utils.py:170: RuntimeWarning: Couldn't find ffmpeg or avconv - defaulting to ffmpeg,
but may not work
warn("Couldn't find ffmpeg or avconv - defaulting to ffmpeg, but may not work", RuntimeWarning)
WARNI [langchain_community.utils.user_agent] USER_AGENT environment variable not set, consider setting it to identify your requests.

OPEN WEBUI
```



6. Problem Statement

The local server (localhost) cannot be accessed from external networks because:
localhost is restricted to the host machine
routers block incoming connections by default
firewalls prevent external access

7. Solution: Using ngrok

ngrok creates a secure tunnel from the local system to the internet, allowing external users to access the local server through a public URL.

8. Installation and Configuration of ngrok

Step 1: Install ngrok

Download from: <https://ngrok.com>

Step 2: Add Authentication Token

ngrok config add-authtoken YOUR_AUTH_TOKEN

Step 3: Start Tunnel

ngrok http 8080

Step 4: Public URL Generated

Example:

<https://abcd-1234.ngrok-free.app>

This URL can be shared with others to access Open WebUI.

9. Commands Used

Virtual Environment Setup

```
python -m venv openwebui-env
```

```
openwebui-env\Scripts\activate
```

Open WebUI Setup

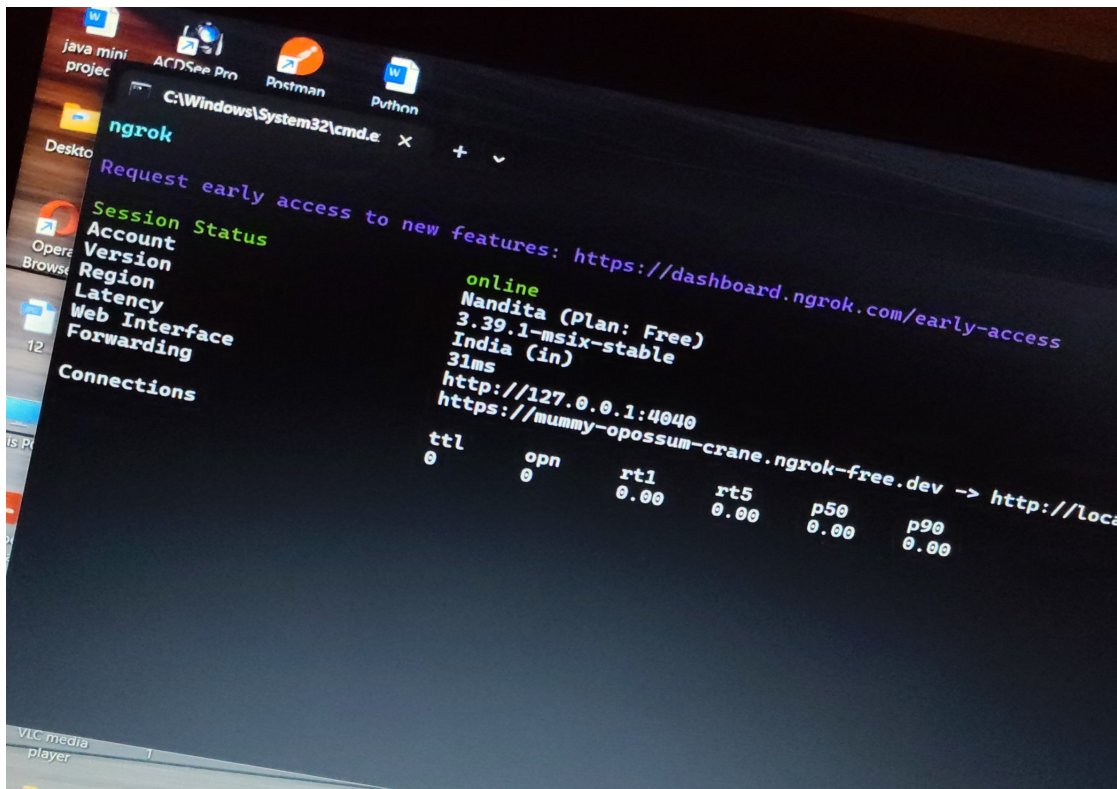
```
pip install open-webui
```

```
open-webui serve
```

ngrok Setup

```
ngrok config add-authtoken YOUR_AUTH_TOKEN
```

```
ngrok http 8080
```

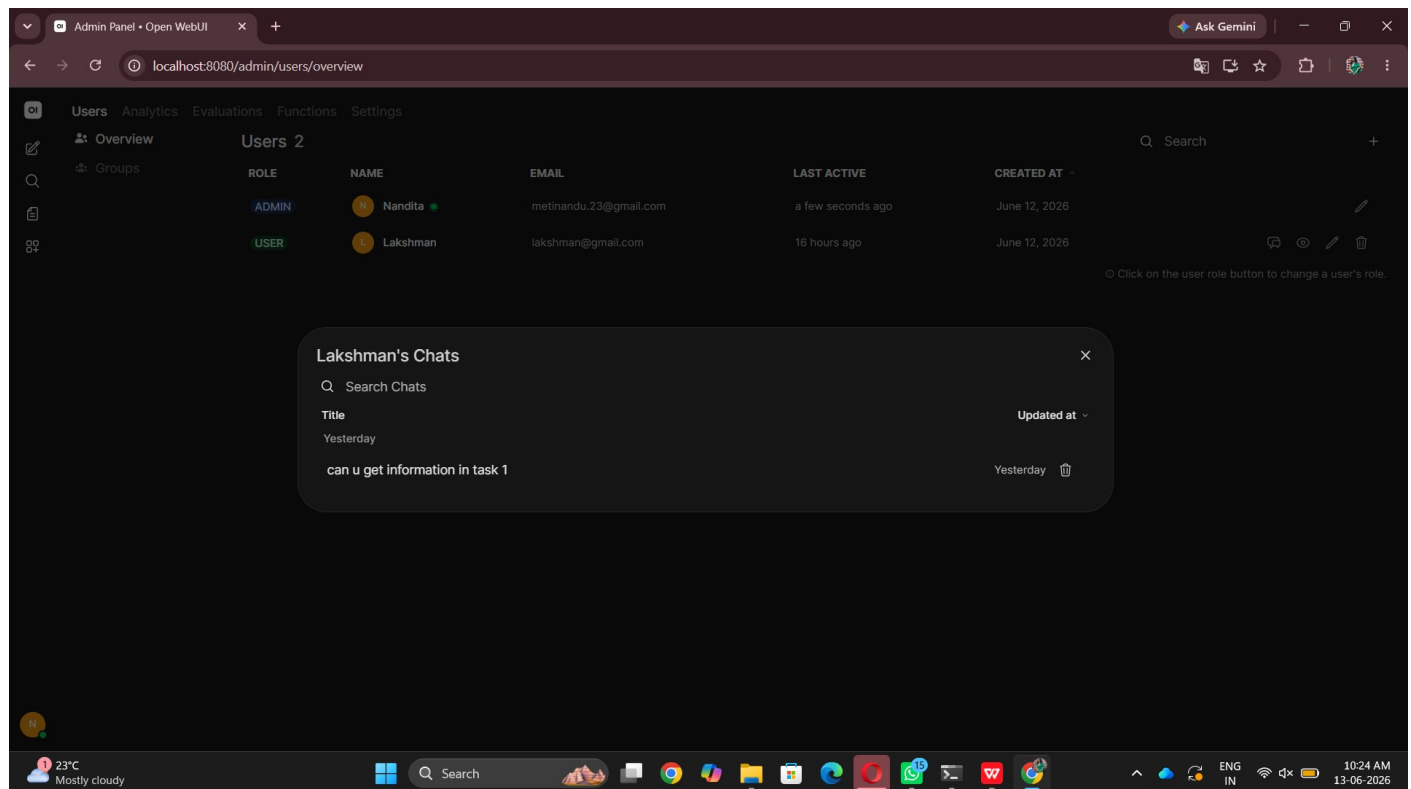
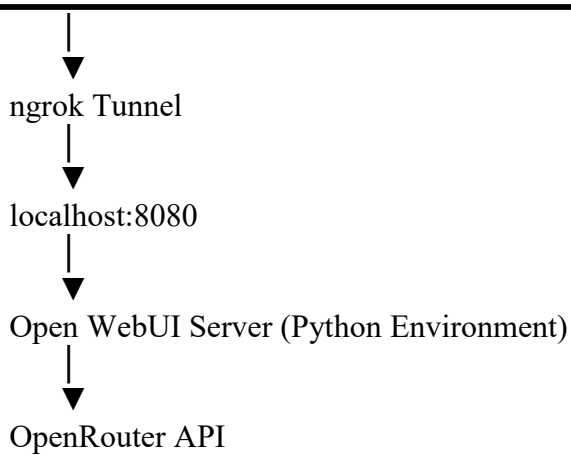


10. Architecture Diagram

External User



ngrok Public URL



11. Advantages

- Easy setup without router configuration
- Secure HTTPS access
- Works on any internet connection
- Useful for testing and demonstrations
- No need for static IP or port forwarding

12. Limitations

- Public URL changes every restart (free version)
- Requires internet connection
- Not suitable for production deployment
- Free plan has usage limitations

13. Alternative Methods

- Port Forwarding in Router
- Cloudflare Tunnel
- Tailscale VPN
- Reverse Proxy (Nginx/Caddy)

14. Conclusion

This project demonstrates how a locally hosted Python-based Open WebUI application can be accessed externally using ngrok. The setup allows secure remote access without modifying router settings or network configuration.

The combination of Python environment, Open WebUI, and ngrok provides an efficient and simple method for sharing local applications over the internet for testing and educational purposes.